

Data Processing and Performance Evaluation

Feature Scaling

- Example:
 - Temperature (°C) $\approx$ 15–40
 - Rainfall (mm) $\approx$ 0–300
- The gradient is proportional to the feature value.
- Large feature values $\rightarrow$ Large gradient
 $\rightarrow$ Larger weight update
- Learning gets over-reliant on Rainfall

$$\hat{y} = w_1x_1 + w_2x_2 + b$$

$$L = \frac{1}{n} \sum (y - \hat{y})^2$$

$$\frac{\partial L}{\partial w_j} = -\frac{2}{n} \sum x_j (y - \hat{y})$$

Feature Scaling

- Standardization

$$x' = \frac{x - \mu}{\sigma}$$

- μ and σ are mean and standard deviation respectively

- Min-Max Scaling

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

- x_{min} and x_{max} are minimum and maximum values respectively

Missing Value Imputation

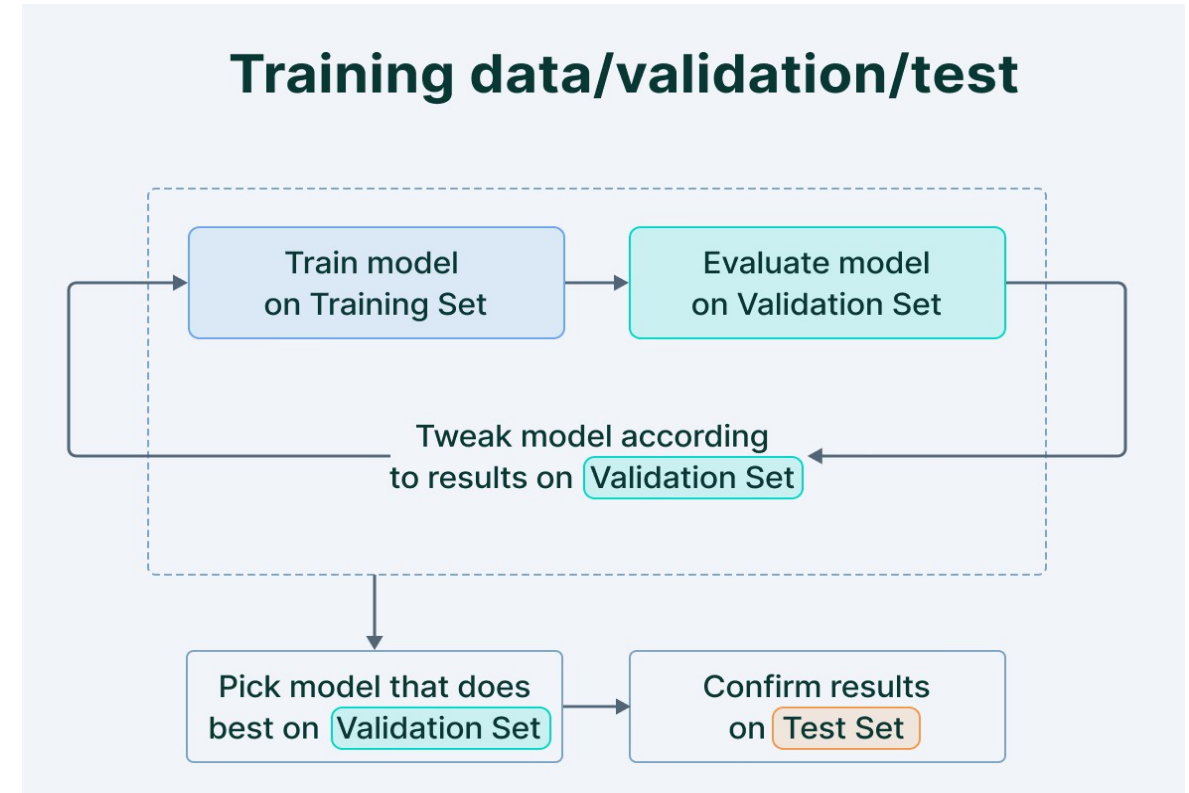
- Imputation: Process of turning incomplete samples into complete samples
- Replace missing values with
 - Mean
 - Minimum
 - Median
 - etc
- ML-based imputation techniques
 - Train a model to predict missing values
 - Ex: KNN-Imputer
- Iterative imputation
 - EX: MICE
- Many more

Data Augmentation

- Noise Addition
- Transformation
 - Rotate
 - Scale
 - Translate, etc
- Synthetic Sample Generation
 - Handle Imbalance
 - SMOTE, Generative Modeling

Training, Validation, and Test Split

- ❑ Training Set
 - Used to train the models
 - Update model parameters
- ❑ Validation Set
 - Used for model selection and hyper-parameter tuning
- ❑ Test Set
 - Report model performance on test set
 - Generalization error
- ❑ Avoid Data Leakage
 - ❑ Example: Avoid standardization using entire dataset



Cross Validation

• ❖ Hold-Out Cross Validation

1. Randomly split D into D_{train} and D_{val}
2. D_{val} is called the hold-out cross validation set
3. Train each model M_i on D_{train} only, to get some hypothesis h_i
4. Select the hypothesis h_i that had the smallest error on the hold out cross validation set
 - Also referred to as the validation error

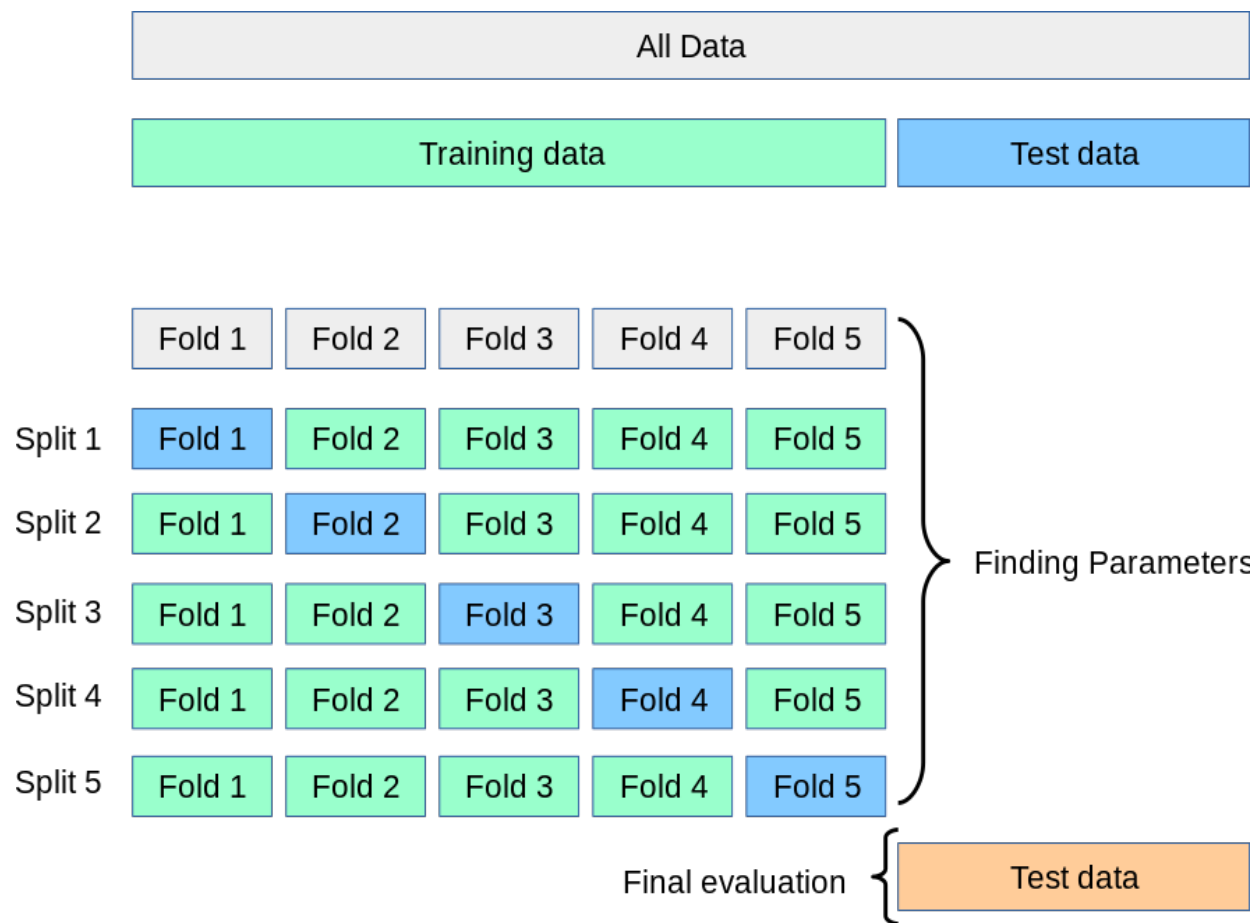
Cross Validation

- ❖ When data is scarce
 - Cannot hold-out the validation set
- ❖ k-Fold Cross Validation

Cross Validation

❖ k-Fold Cross Validation

- Randomly split D into k disjoint subsets
- Treat each split as the validation set and the remaining as the train set
- Pick the model with the lowest estimated generalization error
- Retrain that model on the entire training set D
- The resulting hypothesis is then output as our final answer



Cross Validation

❖ k-Fold Cross Validation

- Randomly split D into k disjoint subsets of N/k training examples
- Let's call these subsets $D_1, \dots, D_k$.
- For each model M_i , we evaluate it as follows:
 - For $j = 1, \dots, k$
 - Train the model M_i on all the data except D_j to get hypothesis h_{ij}
 - Test the hypothesis h_{ij} on D_j , to get the validation error from h_{ij}
 - The generalization error of model M_i is the average validation error of all h_{ij}
- Pick the model M_i with the lowest estimated generalization error
- Retrain that model on the entire training set D .
- The resulting hypothesis is then output as our final answer.

Cross Validation

- ❖ Stratified k-Fold Cross Validation

- Randomly split D into k disjoint subsets of N/k training examples
 - Each fold preserves the class proportions of the full dataset

Cross Validation

❖ Leave-One-Out Cross Validation

- k-Fold Cross Validation where $k=N$
- Leave one sample out as validation set

Performance Metrics

- Accuracy
 - Proportion of examples for which the model produces correct predictions

- If y is the target attribute and $\hat{y}$ is the predicted class,

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(y_i = \hat{y}_i)$$

$$\mathbf{1}(y_i = \hat{y}_i) = \begin{cases} 1, & \text{if } y_i = \hat{y}_i \\ 0, & \text{otherwise} \end{cases}$$

Performance Metrics

- Consider a binary classification task
 - Positive and Negative Classes

-	Originally Positive (1)	Originally Negative (0)
Predicted Positive (1)	True Positive (TP)	False Positive (FP)
Predicted Negative (0)	False Negative (FN)	True Negative (TN)

Confusion Matrix

Performance Metrics

- Accuracy
 - Proportion of examples for which the model produces correct predictions

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

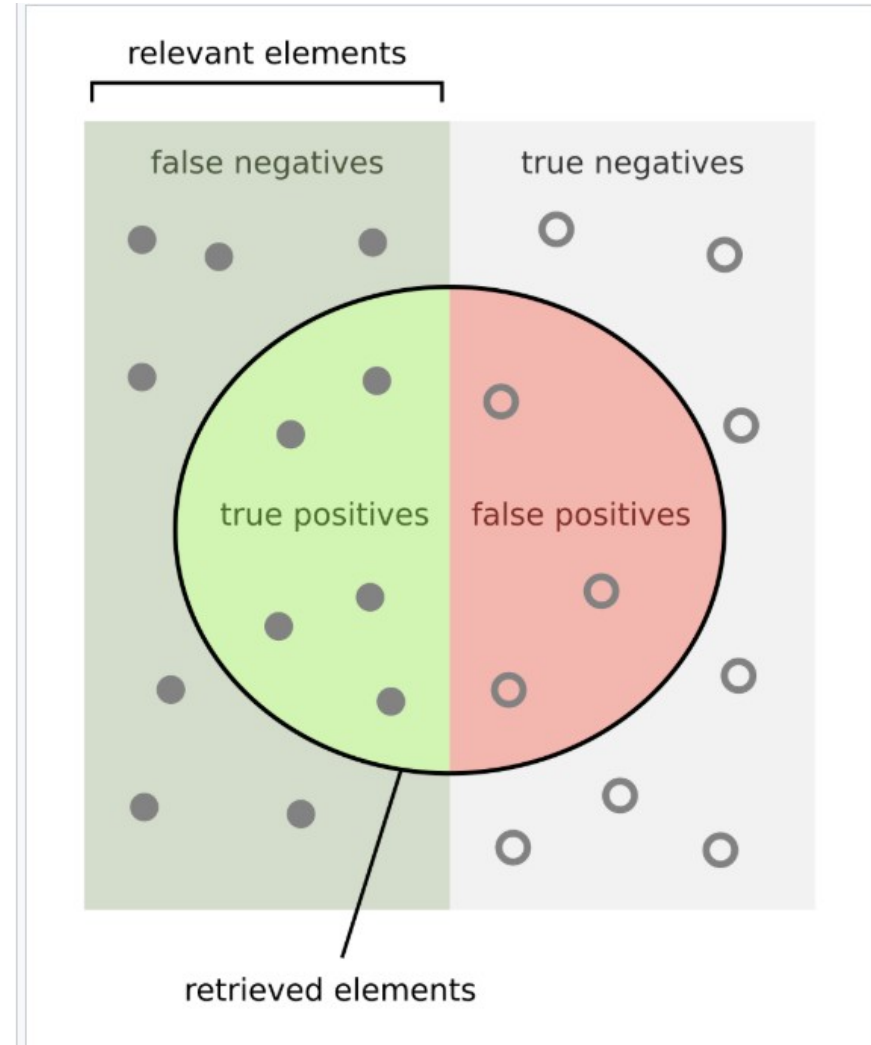
- Does higher accuracy indicate good model?
 - Imbalanced Dataset: What if 90% of the samples are positive and only 10% are negative. [Example: Spam Email Classification](#)
 - Prediction Cost: What if the cost of a False Positive is higher/lower than False Negative? [Example: Recidivism Prediction, Medical Diagnosis](#)

Performance Metrics

- Precision

How many retrieved items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

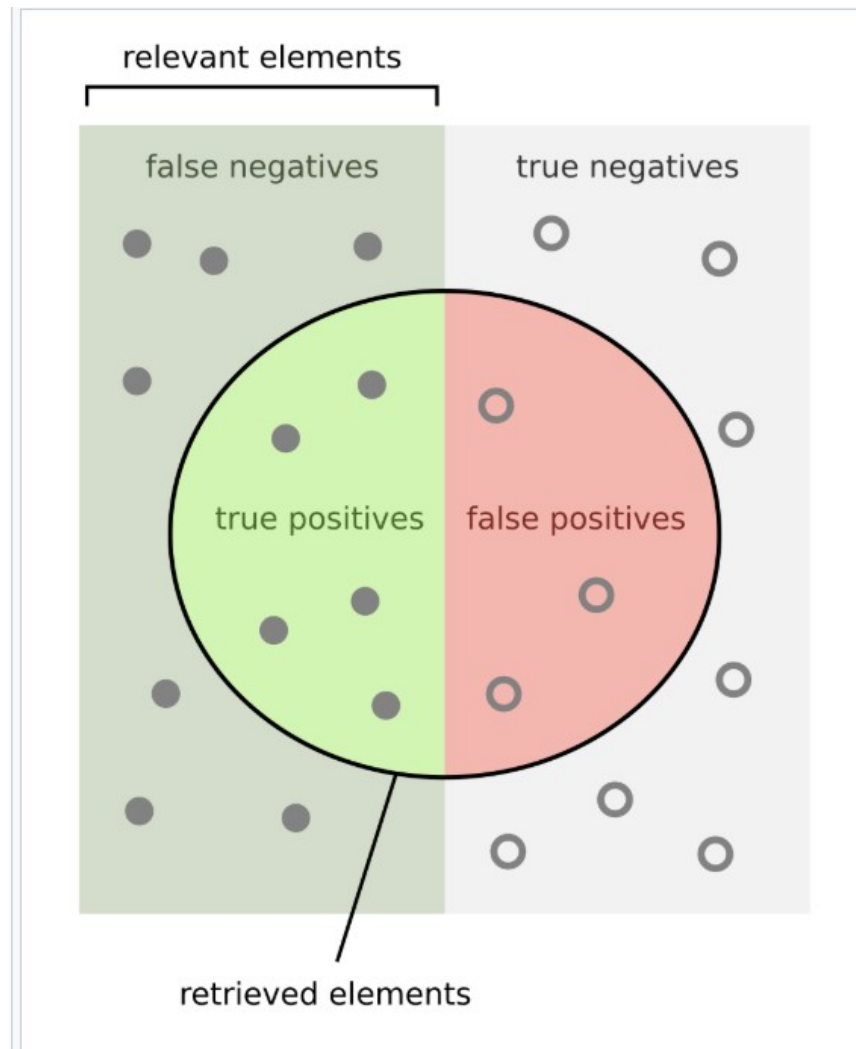



Performance Metrics

- Recall

How many relevant items are retrieved?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

Performance Metrics

- Precision
- Recall

Sources: [\[4\]](#)[\[5\]](#)[\[6\]](#)[\[7\]](#)[\[8\]](#)[\[9\]](#)[\[10\]](#)[\[11\]](#) [view](#) · [talk](#) · [edit](#)

		Predicted condition			
		Total population = P + N	Predicted positive	Predicted negative	
Actual condition	Real Positive (P) ^[a]	True positive (TP), hit ^[b]	False negative (FN), miss, underestimation	True positive rate (TPR), recall, sensitivity (SEN), probability of detection, hit rate, power $= \frac{TP}{P} = 1 - FNR$	Prevalence threshold (PT) $= \frac{\sqrt{TPR \times FPR} - FPR}{TPR - FPR}$
	Real Negative (N) ^[d]	False positive (FP), false alarm, overestimation	True negative (TN), correct rejection ^[e]	False positive rate (FPR), probability of false alarm, fall-out type I error ^[f] $= \frac{FP}{N} = 1 - TNR$	False negative rate (FNR), miss rate type II error ^[c] $= \frac{FN}{P} = 1 - TPR$
	Prevalence $= \frac{P}{P + N}$	Positive predictive value (PPV), precision $= \frac{TP}{TP + FP} = 1 - FDR$	False omission rate (FOR) $= \frac{FN}{TN + FN} = 1 - NPV$	Positive likelihood ratio (LR+) $= \frac{TPR}{FPR}$	Negative likelihood ratio (LR-) $= \frac{FNR}{TNR}$
	Accuracy (ACC) $= \frac{TP + TN}{P + N}$	False discovery rate (FDR) $= \frac{FP}{TP + FP} = 1 - PPV$	Negative predictive value (NPV) $= \frac{TN}{TN + FN} = 1 - FOR$	Markedness (MK), deltaP (Δp) $= PPV + NPV - 1$	Diagnostic odds ratio (DOR) $= \frac{LR+}{LR-}$

Performance Metrics

- F1-Score
 - Achieve balance between precision and recall
 - Harmonic mean of Precision and Recall

$$\text{Precision} = P = \frac{TP}{TP + FP}$$

$$\text{Recall} = R = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2PR}{P + R}$$

$$F_1 = \frac{2TP}{2TP + FP + FN}$$

Performance Metrics

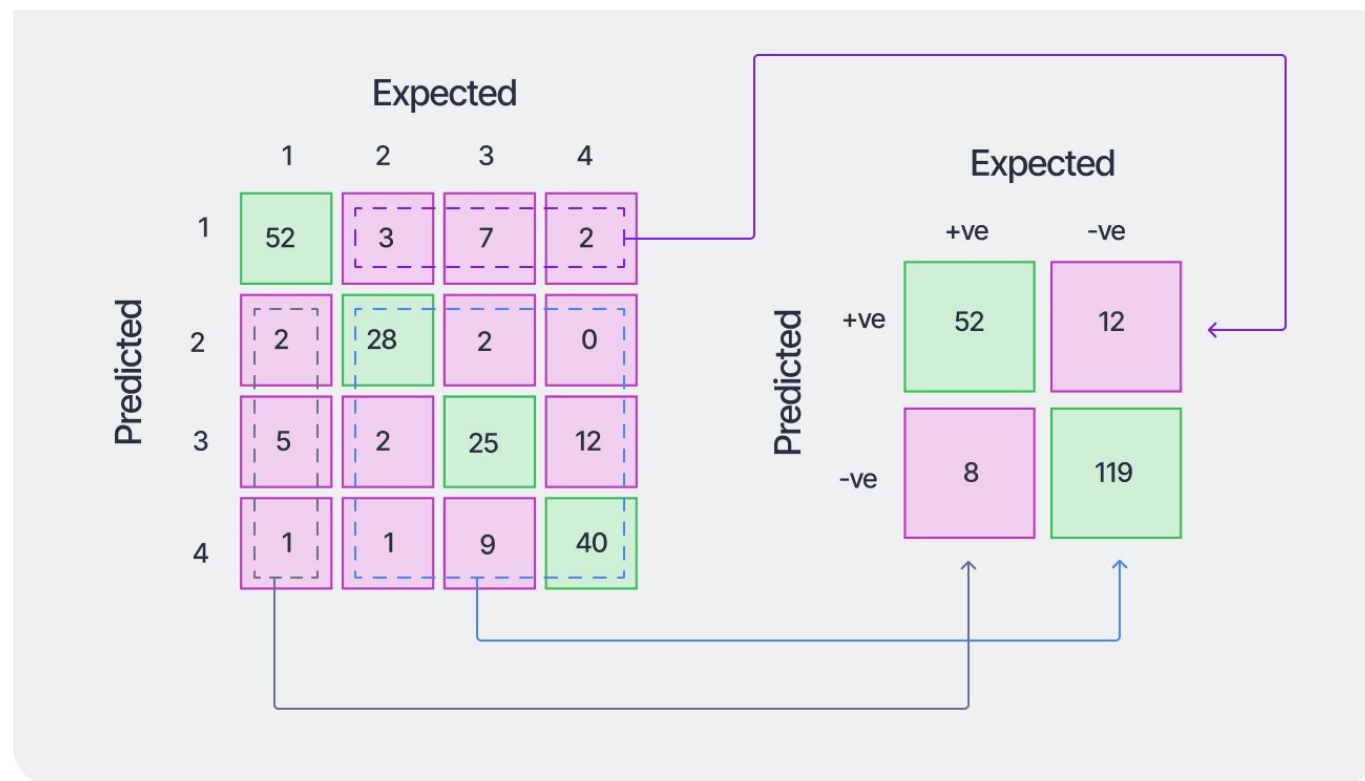
- Performance on multi-class classification task
 - Accuracy

		Expected			
		1	2	3	4
Predicted	1	52	3	7	2
	2	2	28	2	0
	3	5	2	25	12
	4	1	1	9	40

Confusion Matrix

Performance Metrics

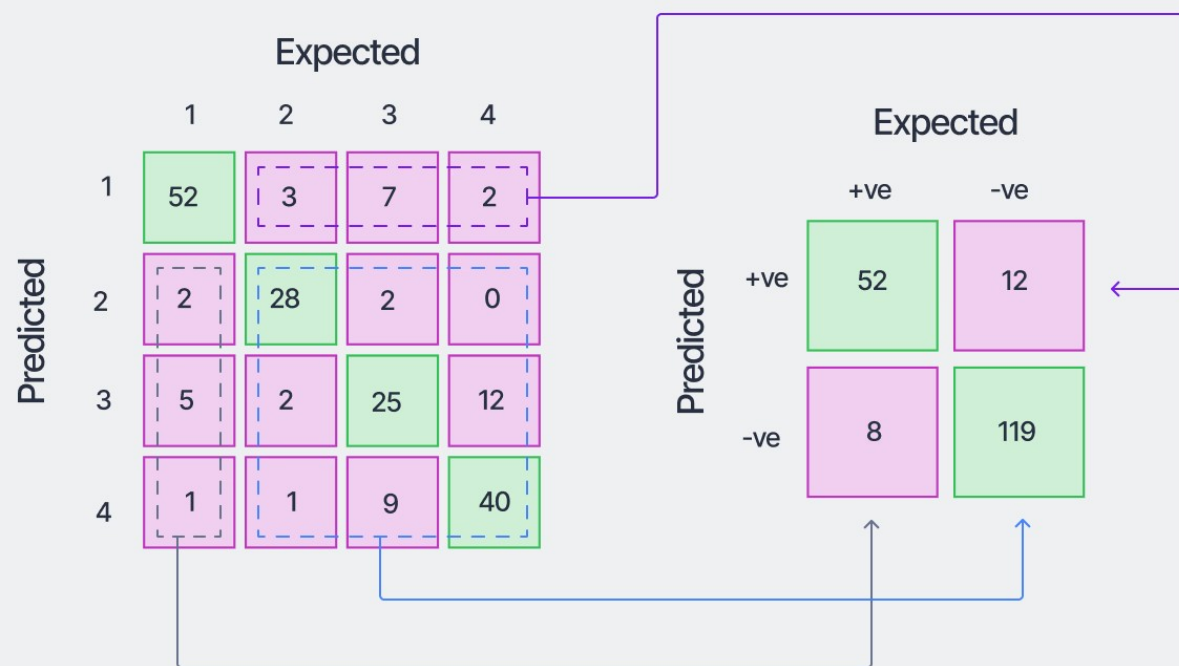
- Performance on multi-class classification task
- One-vs-All
 - One class is considered the positive class
 - The rest are negative class
- Class-wise
 - Accuracy
 - Precision
 - Recall
 - F1-Score



Confusion Matrix

Performance Metrics

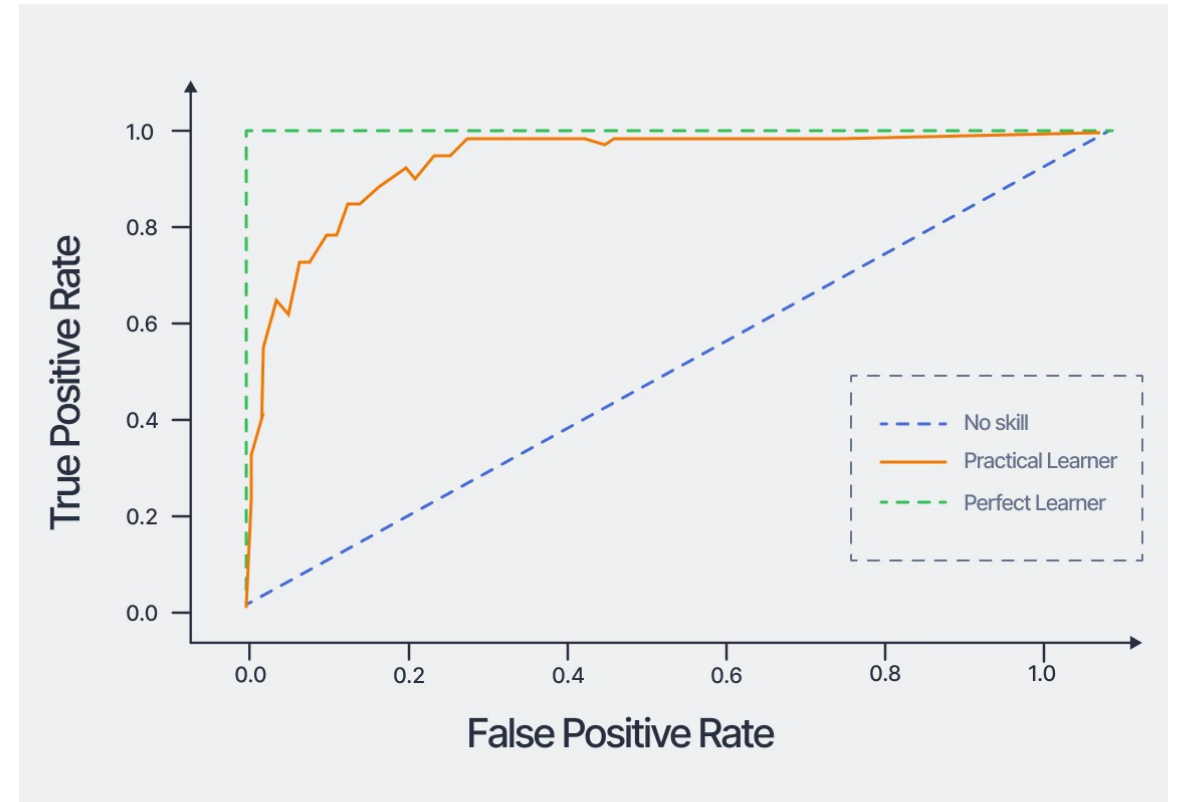
- Performance on multi-class classification task
- Obtain Global Metric
 - Micro-Averaging
 - Macro-Averaging
 - Weighted-Averaging



Confusion Matrix

Area Under Receiver-operating characteristic curve (ROC)

- A threshold invariant metric
- Plot the TPR vs FPR curve by varying the prediction threshold
- Report the Area Under the Curve (AUC)
- High AUC means the model usually assigns higher scores to positive examples than to negative ones.



References

- CS229 Lecture Note
 - Cross Validation: Section 9.3
- Performance Metrics
 - https://en.wikipedia.org/wiki/Precision_and_recall
 - <https://www.v7labs.com/blog/confusion-matrix-guide>
 - <https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc>

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